Operating Systems
Orange Problem - Unit 4

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4B CSE

GitHub Link:
https://github.com/anu-9t/PES2UG24CS073-pes-vcs.git

Screenshot 1A: Output of ./test_objects showing all tests passing

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ rm -rf .pes
make test_objects
./test_objects
make: 'test_objects' is up to date.
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

Screenshot 1B: find .pes/objects -type f showing the sharded directory structure

```
All Phase 1 tests passed.
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

Screenshot 2A: Output of ./test_tree showing all tests passing

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ make test_tree
./test_tree
make: 'test_tree' is up to date.
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

Screenshot 2B: Pick a tree object from find .pes/objects -type f and run xxd .pes/objects/XX/YYY... | head -20 to show the raw binary format

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ xxd $(grep -Rl "tree" .pes/objects | head -n 1) | head -20
00000000: 7472 6565 2034 3900 3130 3036 3434 2068  tree 49.100644 h
00000010: 656c 6c6f 2e74 7874 0066 2246 63d2 3e6f  ello.txt.f"Fc.>o
00000020: 4d9d e9e2 c7e6 d876 4305 a92a 3830 a1a5  M......vC..*80..
00000030: 2d3d 5f4a a800 7b5c 39                   -=_J..{\9
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

Screenshot 3A: Run ./pes init, ./pes add file1.txt file2.txt, ./pes status — show the output

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ rm -rf .pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  commit.h
  untracked:  tree.h
  untracked:  .gitignore
  untracked:  test_objects
  untracked:  test_tree.c
  untracked:  Makefile
  untracked:  commit.c
  untracked:  README.md
  untracked:  object.c
  untracked:  test_objects.c
  untracked:  index.c
  untracked:  test_tree
  untracked:  pes.c
  untracked:  pes.h
  untracked:  index.h
  untracked:  test_sequence.sh
  untracked:  tree.c
```

Screenshot 3B: cat .pes/index showing the text-format index with your entries.

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776459862 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776459862 6 file2.txt
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

Screenshot 4A: Output of ./pes log showing three commits with hashes, authors, timestamps, and messages.

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ rm -rf .pes file1.txt file2.txt hello.txt bye.txt
./pes init

echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: b3c6f676c372... Initial commit
Committed: 5221462e8420... Add world
Committed: 4334bfc0f418... Add farewell
commit 4334bfc0f418981534161c0c81ec9d7f90fd545ed69867b73c51ef54a7de0127
Author: Anupam Tiwari <PES2UG24CS073>
Date:   1776460075

    Add farewell

commit 5221462e84203fe89651587dca399bf94857277a6e12da7df274893d1b71729f
Author: Anupam Tiwari <PES2UG24CS073>
Date:   1776460075

    Add world

commit b3c6f676c372a943fdd94ef992c5318e688ce34414d60ea958201448b63b4381
Author: Anupam Tiwari <PES2UG24CS073>
Date:   1776460075

    Initial commitQ

happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

Screenshot 4B: find .pes -type f | sort showing object store growth after three commits.

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/43/34bfc0f418981534161c0c81ec9d7f90fd545ed69867b73c51ef54a7de0127
.pes/objects/52/21462e84203fe89651587dca399bf94857277a6e12da7df274893d1b71729f
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/b3/c6f676c372a943fdd94ef992c5318e688ce34414d60ea958201448b63b4381
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

Screenshot 4C: cat .pes/refs/heads/main and cat .pes/HEAD showing the reference chain.

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
4334bfc0f418981534161c0c81ec9d7f90fd545ed69867b73c51ef54a7de0127
ref: refs/heads/main
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

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Screenshot: Full integration test (make test-integration)

```
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: dbd5f26942b5... Initial commit

Log after first commit:
commit dbd5f26942b586ac7171a2cd7264a547f155c4a338ec1154d339cab7bb339194
Author: Anupam Tiwari <PES2UG24CS073>
Date:   1776460525

    Initial commit


--- Second Commit ---
Committed: a41989ab58da... Update file.txt

--- Third Commit ---
Committed: c6bd142eaeef... Add farewell

--- Full History ---
commit c6bd142eaeef37a6aec21f0d0e367695dbcf7633a6e893416ac27e85e4c14acc
Author: Anupam Tiwari <PES2UG24CS073>
Date:   1776460525
```

```
--- Third Commit ---
Committed: c6bd142eaeef... Add farewell

--- Full History ---
commit c6bd142eaeef37a6aec21f0d0e367695dbcf7633a6e893416ac27e85e4c14acc
Author: Anupam Tiwari <PES2UG24CS073>
Date:   1776460525

    Add farewell

commit a41989ab58da0a175517787c219a740a8d7eaacb47a946a243b38a4612231caf
Author: Anupam Tiwari <PES2UG24CS073>
Date:   1776460525

    Update file.txter auth!

commit dbd5f26942b586ac7171a2cd7264a547f155c4a338ec1154d339cab7bb339194
Author: Anupam Tiwari <PES2UG24CS073>
Date:   1776460525

    Initial commitQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
c6bd142eaeef37a6aec21f0d0e367695dbcf7633a6e893416ac27e85e4c14acc

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/a4/1989ab58da0a175517787c219a740a8d7eaacb47a946a243b38a4612231caf
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/c6/bd142eaeef37a6aec21f0d0e367695dbcf7633a6e893416ac27e85e4c14acc
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/db/d5f26942b586ac7171a2cd7264a547f155c4a338ec1154d339cab7bb339194

=== All integration tests completed ===
happy@happy-INBook-X1:~/PES2UG24CS073-pes-vcs$
```

Phase 5: Branching and Checkout

Q5.1:

To implement checkout, I'd first update .pes/HEAD to point to the new branch file. Then, I'd sync the working directory by reading the target branch's tree and overwriting or deleting local files to match that snapshot. The complexity comes from the recursive nature of the filesystem and the need to protect "dirty" or uncommitted files so I don't accidentally wipe out my own work during the transition.

Q5.2:

I'd just compare the hashes in my current .pes/index against the hashes in the target branch's root tree. If I've modified a file or staged something, those hashes won't match. If that same file differs in the branch I'm switching to, I'd flag it as a conflict and block the checkout to prevent data loss.

Q5.3:

In a detached state, .pes/HEAD points directly to a commit hash instead of a branch. New commits are saved normally in .pes/objects, but they become "orphaned" as soon as I switch away since no branch reference tracks them. To get them back, I'd just find the orphaned hash and manually create a new branch file in .pes/refs/heads/ to "anchor" it.

Phase 6: Garbage Collection

Q6.1:

I'd use a "mark-and-sweep" algorithm. Starting from the active refs and HEAD, I'd walk back through the commit history and all linked trees/blobs, storing "reachable" hashes in a **Hash Set** for O(1) lookup

efficiency. Then, I'd delete everything in .pes/objects/ that isn't in that set. For 100,000 commits, I'd visit at least 200,000 objects (commits + root trees), though deduplication means I only process unique content once.

Q6.2:

It's a race condition. If I run pes add, a blob is created but isn't yet linked to a commit. If GC runs at that exact second, it sees an "unreachable" blob and deletes it, causing my final commit to fail. Real Git fixes this by checking the file's modification time (mtime) and ignoring any unreachable objects created recently—usually within a 14-day grace period.